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Benford's law and intraday microstructure anomalies: Forecasting market movements with high-frequency data

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ABSTRACT

Benford's Law has been widely used to detect data irregularities across various domains, including finance. While prior work (Hamida et al., 2024) demonstrated its predictive relevance for daily stock returns, this study extends that framework to high-frequency intraday data, offering a new perspective on short-term market dynamics. Specifically, we apply Benford's Law to the first-digit distributions of intraday stock returns, trading volume, and trade durations, two microstructural variables never explored in Benford-based financial research. Using data from twenty Euronext Paris-listed companies, we uncover significant deviations from Benford's distribution that are particularly pronounced during anomalous market periods. These deviations are then incorporated into both linear and switching regime models to assess their predictive impact. Our findings reveal that Benford-derived indicators are conditionally informative, showing predictive value primarily during anomalous market periods, whether driven by intentional manipulation or by significant, non-fraudulent irregularities. These indicators function as effective early-warning signals for abnormal intraday dynamics. The study advances both forecasting and forensic finance by integrating a diagnostic tool into real-time predictive modeling in high-frequency trading environments.

1. Introduction

The predictability of stock market returns is a paramount concern in the financial sector, as more accurate forecasts translate into increased profitability from trading strategies (Fletcher and Shawe-Taylor, 2013; Kercheval and Zhang, 2015). Central to this quest for predictability is the concept of financial market efficiency, which posits that stock returns fully incorporate all pertinent information and, therefore, are not readily forecasted. Eugene Fama was the pioneer in introducing the idea of "efficient markets" in his seminal work, wherein he argued that asset prices are determined by available information (Fama, 1963, 1965, 1970, 1995). The quality of predictions is influenced by two major factors: the choice of methodology for training prediction models and the selection of input variables (predictors). Notably, Qi (1999) examined the predictability of Standard and Poor's 500 index returns by applying a linear regression model and a nonlinear neural network (NN) model. Results of his work show that the nonlinear neural network (NN) model fits the data better than the linear model in the sample, and also provides fairly accurate forecasts out of the sample. Similarly, Qian and Gao (2017) suggested that classical machine learning methods outperform traditional models in

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the precision of financial time series predictions. Additionally, Matias and Reboredo (2012) analyzed the predictability of S&P 500 intraday returns by comparing linear and nonlinear models. Their findings showed that nonlinear models outperformed linear ones based on statistical and economic measures. However, more flexible nonlinear models, like support vector machines and artificial neural networks, did not consistently surpass other nonlinear approaches.

Recently, the prediction of high-frequency trading (HFT) data has gained significance for practitioners, offering the potential for substantial annual returns in intraday trading strategies. However, concerns about high-frequency market manipulation have emerged. Market manipulation distorts prices, leading to reduced market efficiency and economic losses from misallocated resources (Pirrong, 1995). To combat this, Benford's law (Benford, 1938) has become a widely employed method to detect manipulated data. In the field of management, Lin et al. (2018) applied Benford's Law to data from 501 Taiwanese "Fat Cat" companies to address the limitations of accrual models in detecting earnings manipulation. Their findings indicate that earnings management is prevalent among most listed companies in Taiwan. Similarly, in banking, Parnes (2022) examined banks' off-balance sheet items, revealing significant manipulation through excessive rounding and frequent use of digits 0 and 5, although regulatory frameworks like the Basel Accords have helped mitigate some of these irregularities. In financial analysis, Benford's law is frequently used by researchers to scrutinize the distribution of the leading digits in stock prices. For example, Riccioni and Cerqueti (2018) explored the applicability of Benford's law on all the indexes listed in international financial markets by analyzing stock prices and volumes. Using the Chi-square test for both the distributions of the first and second meaningful digits, they identified violations and extracted insights related to economic shocks and historical events. Departures from the expected distribution can act as an early indication of abnormal market behavior or possible fraudulent activities, both of which might affect future price fluctuations. Thus, this analytical method can forecast increased volatility, which often leads to noticeable decreases in stock values.

Market inefficiencies are a known feature of financial systems, yet certain anomalies within high-frequency trading data may reflect deliberate or systemic manipulation rather than random deviations. The ability to detect such anomalies in real time, before they translate into significant price movements, offers a strategic edge for traders, regulators, and analysts. In this study, we extend the application of Benford's Law, a statistical tool for identifying irregularities in numerical datasets, to the microstructure level by examining the first-digit distribution of intraday stock returns, trading volumes, and trade durations for 20 companies listed on Euronext Paris. This represents a novel contribution, as prior research has largely focused on daily or aggregated financial data. Using statistical tests, including the chi-squared test, the MAD test, and the Z-statistics test, to quantify deviations from the expected Benford distribution, we assess whether these irregularities can serve as predictive signals for intraday stock returns. Building on the forecasting framework of Hamida et al. (2024), we integrate Benford-derived variables into linear and smooth transition autoregressive (STAR) models. Our findings reveal that Benford's Law metrics offer limited insight under normal market conditions, when pricing dynamics generally reflect efficient behavior. However, their predictive power increases significantly in contexts characterized by market inefficiencies or the presence of intentional or unintentional irregularities. This effect is especially pronounced in high-frequency trading environments, where manipulative practices tend to be more nuanced and short-lived.

Unlike previous studies, which focused on daily stock returns, the present work investigates Benford deviations within an intraday, high-frequency environment, allowing us to uncover dynamics that cannot emerge at lower frequencies. Because markets have significantly less time to adjust within very short intervals, intraday prices tend to exhibit lower informational efficiency than daily prices, which reflect a longer adjustment period. This lower efficiency at high frequencies allows short-lived distortions and numerical irregularities, often invisible in daily data, to emerge more clearly, thereby offering deeper insight into the mechanisms underlying Benford deviations. Therefore, the use of intraday data is essential as it preserves the short-lived fluctuations that characterize microstructure behavior, such as temporary liquidity imbalances, rapid order cancellations, and manipulation attempts, that are typically smoothed out when observations are aggregated into daily indicators. By working at the temporal scale at which these events occur, we are able to detect Benford irregularities in real time, highlighting their potential as an operational early-warning tool for market supervision and short-horizon risk monitoring. In addition, the study incorporates microstructure variables that extend the current Benford-based literature, especially trade duration, an inherently intraday measure that captures the time gap between consecutive trades. Unlike trading volume, which can be meaningfully aggregated at the daily level, trade duration cannot be reconstructed from daily data and therefore opens a new forensic dimension. Its first-digit distribution provides direct insight into trading intensity, algorithmic activity, and liquidity stress, elements that are invisible in daily-frequency analyses. Deviations in duration-based Benford patterns reveal types of microstructural disturbances, including strategic accelerations or slowdowns in trade execution, that leave no trace once data are aggregated. Finally, our empirical results show that Benford deviations intensify precisely during short, abnormal trading episodes. Overall, this paper contributes to the literature on financial forensics and market microstructure by demonstrating that Benford-based indicators can detect microstructural irregularities, support real-time detection of abnormal trading behavior, and potentially enhance intraday trading strategies when combined with conventional technical tools such as the Relative Strength Index, Bollinger Bands, or the Average True Range.

The remainder of this paper is structured as follows: Section 2 provides a review of prior works on the various methods used to identify data manipulation and introduces Benford's law and its application in several fields. Section 3 outlines the dataset used in this study and presents the methodology, along with a brief theoretical background on STAR models. Section 4 reports and discusses the empirical results, and finally, Section 5 offers the conclusion.

2. Literature review

2.1. Stock market manipulation

Manipulation is an issue of importance for both developed and emerging stock markets and has drawn the attention of many researchers. Allen and Gale (1992) and Jarrow (1992) are among the first who tackle this problem. According to these authors, there exist three basic types of stock price manipulation. The first type involves activities that impact the perceived or actual value of assets, known as action-based manipulation. The second type is information-based manipulation, which happens when misleading information or untrue rumors are spread. Finally, traders who engage in trade-based manipulation attempt to impact stock prices by merely buying and selling without changing the firm's value or disseminating misleading information. In light of this, Van Bommel (2003) developed a model for information-based stock market manipulation to analyze empirical data on stock returns and trading volume around rumored events. Hanson et al. (2006) presented an experimental model to investigate whether manipulative traders could distort the prices in a prediction market by spreading false information. At the same time, many studies provide evidence of the manipulation of closing stock prices (Comerton-Forde and Putniņš, 2011; Spatt, 2014). Also, Abrantes-Metz et al. (2012) and Fouquau and Spieser (2015) investigated the manipulation of the London inter-bank offered rate (LIBOR). Using the complete intraday order and trade data of the Korea Exchange (KRX) in a custom data set identifying individual accounts, Lee et al. (2013) found that investors strategically placed spoofing orders which created the impression of a substantial order book imbalance, with the intent to manipulate subsequent prices. Recognizing these challenges, researchers have developed diverse methods for detecting abnormal stock price changes and identifying instances of stock price manipulation. Öğüt et al. (2009) used statistical analyses and artificial neural networks to detect abnormal trading activity that might indicate manipulation in the Istanbul Stock Exchange. Kim and Park (2010) developed a theoretical model suggesting that daily price limits can deter stock market manipulation, particularly in environments with a high risk of such activity. Using the complete intraday order and trade data of the Korea Exchange (KRX) in a custom data set identifying individual accounts, Lee et al. (2013) found that investors strategically placed spoofing orders which created the impression of a substantial order book imbalance, with the intent to manipulate subsequent prices. Eisl et al. (2017) examine the manipulation potential in the ratesetting processes of LIBOR and EURIBOR by analyzing individual bank submissions. Their findings show that alternative fixing methods could substantially reduce the scope for manipulation, highlighting vulnerabilities in existing benchmark mechanisms. In a more recent investigation,(El Mouaaouy and Riepe, 2018) created a manipulation measure that decomposes foreign exchange (FX) rates into several digit positions and investigates the proportion of digit differences between World Markets Company and Reuters benchmarks and Thomson Reuters rates to detect anomalies of misconduct in the FX market. Furthermore, Malenko et al. (2023) proposed a quadrophobia score based on earnings per share across all publicly traded firms to identify firms with aggressive accounting cultures.

For investors and institutions, the compliance of financial data with particular functional distributions may be seen as a decision support system in the fields of data mining and decision analytics. A few noteworthy contributions to this field are worth mentioning. For example, Pietronero et al. (2001) investigated the underlying causes of the uneven distribution of numbers in natural systems and showed Zipf's law's effectiveness in detecting fraud. Similar results are obtained by Huang et al. (2008), who confirmed that Zipf's law can be used by auditors as a tool for fraud detection. In the past thirty years, extensive work has applied Benford's law to analyze data sets and to identify digit patterns that may indicate possible misconduct. In the next subsection, we present a literature overview of Benford method applications.

2.2. Applications of Benford's Law across financial and managerial domains

Benford's law has been extensively used as a tool to identify abnormal behaviors in several data sets ranging from accounting fraud detection (see Grammatikos and Papanikolaou (2021)), taxation (see Mir et al. (2014), Ausloos et al. (2017)), population numbers, natural phenomena, stock market (Karavardar, 2014), and recently COVID-19 cases reports. Berger and Hill (2011) discussed in detail the methodology and basic mathematical grounds of this approach. Nigrini appears to be among the first academics to use Benford's law within accounting data to look for potential fraud (Nigrini, 2017, 2019). Similarly, Bhattacharya et al. (2011) explored the use of this law as a decision support system for fraud detection and for identifying the occurrence of tax evasion. In management, El Mouaaouy (2018) used Benford law to conduct a digit analysis on a sample of 19 sizable German financial conglomerates to identify managerial engagement in the capital allocation process. They discovered that this method can only be used as an indicator of administrative engagement in an allocation outcome based on Pearson's $\tilde{\chi}^2$ test. Hales et al. (2008) suggested Benford's law as a tool for identifying suspect employee-reported data. In Banking, Harb et al. (2023) applied Benford's first- and second-digit tests to investigate the manipulation patterns of the ten largest Lebanese banks. Z-statistic and Chi-square tests are employed to examine deviations from BL frequencies. The findings indicate that banks may have significantly manipulated their capital adequacy, liquidity, and asset quality during and after the pre-financial engineering era. Adam and Tsarsitalidou (2022) employed Benford's law to determine whether there is evidence of data misreporting in the overall number of COVID-19 reported cases across different countries. The deviation of actual data from those predicted by Benford's law was measured by the Mean Absolute Deviation. Results demonstrate that autocratic countries are more likely to underreport COVID-19 instances when using the Instrumental Variable model of Lewbel (2012) and Settler Mortality as an external tool for democracy. Additionally, Barabesi et al. (2018) proposed a hierarchical testing procedure as a pre-requisite for the concrete application of Benford's law methodology in anti-fraud applications for international trade data. Regarding financial markets, Ley (1996) documents that the first digits of the daily returns of the Standard and Poor's Index (S&P) and the Dow Jones Industrial Average Index (DJIA) closely agree with Benford's distribution. More recently, Rauch et al. (2013) suggested Benford's law as an appropriate method to detect LIBOR manipulation. They found that the first digit distribution of the reported 150 LIBOR rates did not follow the expected pattern of decreasing frequency from 1 to 9, but rather showed an excess of 1's and a deficiency of 5's and 6's. Additionally, Cinko (2014) pointed out that the day return of BIST-100 fit Benford's Law. In the study of Nigrini (2015), the author reviewed the strategy of segmenting a population into subsets and examined the applicability of Benford's law on daily returns, daily volumes, expected returns, and abnormal returns. In the study of Vičič and Tošić (2022), the authors proposed Benford's law as a preliminary screening tool to identify abnormal behavior and possibly detect fraud in all transactions on the Ethereum (ETH) network. The results show that all the big blockchain platforms by market capitalization that were not biased by any big scandal or lawsuit conform to Benford's distribution. However, they demonstrated that this method could produce false positives in the form of non-conformity in a cryptocurrency, with no fraudulent reason found. One of the more recent researches involving Benford's law is (Morales et al., 2022). The authors applied Benford's law to a high-volume database of financial transactions in a large state-owned enterprise in the energy sector of Argentina, as a means of searching for possible abnormalities and assessing the effectiveness of internal audit procedures. The empirical results of this work indicate that Benford's law can be used as a tool for fraud detection and improving financial accountability in large organizations. Among this body of studies, Cerqueti et al. (2022) highlighted the role of Benford's law as a financial market risk predictor. The empirical findings of their study reveal that the *p*-value of the test $\tilde{\chi}^2$ against the Benford distribution exhibits some forecasting power for the daily returns of the market index and the risk level using the series of daily returns of the Standard and Poor's 500. Based on these studies, a deviation of the observed digit distribution of the data from Benford's expected frequencies would indicate irregularities and fraudulent activities and could impact future price fluctuations. A significant recent contribution to this field is the research conducted by Sifat et al. (2024). The study investigated potential illegal trading in non-fungible token (NFT) markets by utilizing a thorough analytical framework. This framework included Benford's Law, clustering using Student's t-test, and Pareto-Levy distributions to identify irregularities. Investigating the trading volumes of 50 highly popular NFTs, the authors discovered indications of possible price manipulation, underscoring the need for regulatory interventions. Their conclusions, reinforced by supplementary statistical tests such as the χ^2 test and the Mean Absolute Deviation statistic, highlight the disorderly and uncontrolled characteristics of digital asset markets. This aligns with current research trends in both the cryptocurrency and traditional financial markets. More recently, Hamida et al. (2024) demonstrated that the first digits of daily stock returns from the Euronext Paris and Tunisian stock markets do not conform to Benford's distribution, highlighting the presence of nonlinear dynamics in return prediction. Building on the work of Hamida et al. (2024), this study is based on the theoretical premise that deviations from Benford's expected digit distribution signal potential irregularities or manipulative behaviors in financial markets. Extending this approach to a high-frequency intraday setting, we propose that such anomalies, detected through deviations in returns, trading volumes, and durations, may precede short-term price fluctuations, offering early insight into abnormal market conditions.

3. Materials and methods

3.1. Materials

For the present study, we performed an empirical application on 20 French stocks of different sizes (small-medium enterprises and large enterprises) traded on the French Stock Market (see Table 1). These stocks were randomly sampled without rigorous consideration of the kind of economic impact they have on society. The French market offers a suitable context for our analysis due to its diverse set of companies, active intraday trading, and highly detailed microstructural data, which allows us to examine potential deviations from Benford's Law. For each stock, we have access to intraday data, which was continuously computed from 08:00 to 16:30 GMT. The database was retrieved from the AMF - BEDOFIH European high-frequency database[1] and covers a period starting on 01/03/2022 and ending on 12/30/2022. In these files, among many other information, we extracted the stock trading prices, the trading volume, and the transaction timestamps. The selection of twenty companies was primarily guided by the availability of high-frequency intraday data and the requirement to include firms with sufficient trading activity to ensure reliable analysis of first-digit distributions and microstructure variables (returns, volumes, and trade durations). At the time of data collection, only data for 2022 were accessible. To maintain consistency across the series, we aimed to gather a comparable number of observations for each company. As a result, for some companies, this required collecting data over three to four months, while for others, a full year was necessary to achieve the desired sample size.

[1] The data was obtained via EUROFIDAI Equipex PLADIFES ANR-21-ESRE-0036. More information on the BEDOFIH European High-Frequency financial database can be found at the following link: https://www.eurofidai.org/en/bedofih-database.

Table 1
The twenty stocks from the French Stock Market.

Companies	Stocks	Observations (Prices)	Period of study
AXA	AXA	63,018	01/03/2022 to 01/07/2022
L'Oréal	OR	45,508	01/03/2022 to 01/06/2022
Total Energies	TTE	54,182	01/03/2022 to 01/04/2022
Air France - KLM	AF	30,219	01/03/2022 to 01/06/2022
Alstom	ALO	49,665	01/03/2022 to 01/11/2022
Crédit Agricole	ACA	47,962	01/03/2022 to 01/06/2022
Eurofins Scient	ERF	35,303	01/03/2022 to 01/06/2022
Forsee Power	FORSE	26,077	01/03/2022 to 12/30/2022
Safran	SAF	45,824	01/03/2022 to 01/06/2022
Schlumberger	SLBDS	54,380	01/03/2022 to 12/30/2022
AB Science	AB	22,252	01/03/2022 to 02/28/2022
Altarea	ALTA	38,032	01/03/2022 to 06/17/2022
Atari	ALATA	26,928	01/03/2022 to 03/31/2022
Believe	BLV	23,324	01/03/2022 to 03/16/2022
Genfit	GNFT	25,946	01/03/2022 to 03/11/2022
Neoen	NEOEN	31,268	01/03/2022 to 04/29/2022
Nacon	NACON	31,268	01/03/2022 to 04/29/2022
Poxel	POXEL	21,166	01/03/2022 to 04/13/2022
Voltalia	VLTSA	43,363	01/03/2022 to 03/23/2022
Wavestone	WAVE	42,031	01/03/2022 to 05/31/2022

3.2. Benford's law

In 1938, Frank Benford rediscovered a finding first observed by the astronomer and mathematician Simon Newcomb in 1881 and demonstrated that the digits 1 to 9 are not uniformly distributed and ones occur most often and nines least often as shown in Table 2.

Table 2
Expected first digit frequencies based on Benford's law.

Digit	0	1	2	3	4	5	6	7	8	9
First position	–	30,10%	17,60%	12,50%	9,70%	7,90%	6,70%	5,80%	5,10%	4,60%

According to Benford (1938), the distribution of the first digits n of observed data can be approximated by the following relation:

$$P(\text{first digit} = n) = \log_{10}(n+1) - \log_{10}(n) = \log_{10}\left(1 + \frac{1}{n}\right), \quad n = 1, \dots, 9 \tag{1}$$

The quantity $P(n)$ is proportional to the space between n and $n+1$ on a logarithmic scale. To verify that a dataset adheres to Benford's law, several criteria must be met, as established by Benford (1938) and elaborated by Drake and Nigrini (2000). Firstly, the dataset should include numbers with at least three digits. If a number begins with a zero or a comma, the next natural digit is considered the first. Additionally, the dataset should reflect the sizes of similar phenomena; for instance, combining market values with the net income of listed companies in one dataset is inappropriate. The dataset should not have any restrictions on minimum or maximum values, meaning no data points should be excluded. The dataset's mean should be larger than the median, suggesting a right-skewed distribution, and it should ideally have a long right tail (Vičič and Tošić, 2022). Besides these widely recognized criteria, there are variations in the strictness of Benford's Law application. One confusion is whether the three-digit requirement means numbers must be at least 100 or simply have three digits, such as 0.05. Benford noted that zeros or commas preceding the first natural number should be ignored (e.g., 0.015 is considered to have 1 as the first digit and 5 as the second).

3.3. Statistical tests for assessing conformity to Benford's Law

To evaluate the compliance of each time series to Benford's distribution, several statistical tests commonly discussed in the literature can be applied, including the chi-square goodness-of-fit test, the Mean Absolute Deviation (MAD) measure, and the Z-statistic test. The chi-square test, widely used in prior studies such as (Durtschi et al., 2004), evaluates the overall discrepancy between observed and theoretical digit frequencies. The chi-squared statistics, employed for this assessment, are mathematically represented as follows:

$$\tau_j = \sum_{i=1}^{9} \left(\frac{N_i - np_i}{np_i} \right)^2 \sim \chi^2_{(8)},$$

where τ_j: is the chi-square statistics of the asset j; N_i: is the actual count of the significant digit i; np_i: is the expected count of the significant digit i (p_i is the theoretical probability of obtaining the significant digit i and n is the total number of significant digits); The number of degrees of freedom is $K - 1$ (K represents the number of significant digits ($K = 9$)). Under the null hypothesis, the sample satisfies Benford's Law.

The Mean Absolute Deviation (MAD): this measure, which captures the average absolute deviation between actual and expected proportions, reflects relative differences and is less sensitive to sample size (Nigrini and Miller, 2007).

The Z-statistic test: the Z-statistic quantifies the magnitude of deviation for each digit by comparing observed and Benford-expected frequencies. Larger Z-values indicate greater divergence, and values exceeding 1.96 typically signal statistical significance at the 5% level, leading to rejection of the null hypothesis of conformity (Sadaf, 2016; Alali and Romero, 2013).

3.4. Explanatory variables and Benford's variables

From the resulting clean-depurated source data, three time series were constructed: the log-return time series, the log-volume time series, and the duration time series. The log returns are calculated from one-second intraday prices using the formula below:

$$R_t = \ln\left(\frac{P_t}{P_{t-1}}\right),$$

where P_t refers to the one-second price of the asset in the current time t, and P_{t-1} denotes the one-second price of the same asset in the previous time $t-1$. As we analyzed one-second stock returns, we observed a significant occurrence of zero returns. Consequently, a deliberate decision was made to exclude these zero returns from our analysis, focusing exclusively on non-zero returns. The resulting dataset, with details presented in Table 3, reflects our refined observations. To ensure stationarity the log volumes are obtained using the following relation:

$$V_t = \ln\left(\frac{V_t}{V_{t-1}}\right),$$

where V_t presents the one-second volume of the asset in the current time t, and V_{t-1} is the one-second volume of the same asset in the previous time $t-1$. In handling high-frequency data, it becomes imperative to account for the temporal intervals separating individual transactions meticulously. In the present investigation, a crucial step in this process involves the conversion of each datetime value into Unix time, executed through the "time.mktime(x.timetuple())" python function. This computational transformation provides us with a quantification of the elapsed time in seconds, reckoned from the epoch date of January 1, 1970. Subsequently, the temporal span denoted as D_t, which signifies the duration in seconds between two consecutive transactions, is determined as follows:

$$D_t = \text{Unix\_time}(t) - \text{Unix\_time}(t-1)$$

As detailed Section 1, the present study endeavors to enhance the accuracy of one-second stock return predictions by incorporating various explanatory factors, such as asset returns, asset volume, temporal duration between consecutive transactions, and the focal point of this study, Benford's variables calculated on each factor. The central objective of our inquiry is to ascertain the extent to which Benford's variable imparts significant information on intraday stock returns. To build Benford's variable from log returns, log volumes, and duration, we divided in the first step the chi-squared test statistics, which is calculated on the individual window level $[(t-u+1, t)]$, by the sample size N:

$$c = \sum_{i=1}^{9}\left(\frac{(h_i - p_i)}{p_i}\right)^2,$$

where c is a measure of deviation, h_i is the observed relative frequency, and p_i is the relative frequency expected by Benford. This technique enables us to avoid the effect of larger samples leading to higher deviations even if there are no changes in the relative frequencies being observed (Leemis et al., 2000). In the second step, Benford's variable, denoted by B_t, is given by: $B_t = \ln(c)$. Benford's variables are calculated on the individual window level $[(t-u+1, t)$ with $u = 200$ consecutive transactions$]$. Subsequently, we derived three additional time series, namely the intraday asset return divergence from Benford's distribution denoted as Br_t, the one-second asset volume deviation from Benford's distribution denoted as Bv_t, and the time interval between successive transactions divergence from Benford's distribution denoted as Bd_t.

3.5. Methods

Prior academic studies focused on the linear relationship between stock returns and available financial and economic information. They sought to forecast stock returns based on linear regression methods (Litzenberger and Ramaswamy, 1982; Yao et al., 2005). Indeed, and due to the inadequacy of conventional linear models, an increasing body of empirical studies emphasizes the importance of the nonlinear framework. In this work, we apply both linear and nonlinear time series models.

3.5.1. Linear model

Before delving into the application of nonlinear models, we investigate the presence of a linear correlation between stock returns and Benford's variables through the application of a first-order linear autoregressive (AR) model. In pursuit of this objective, we assess two distinct models, as delineated in Eqs. (2) and (3).

$$R_t = a_{0,0} + a_{0,1}R_{t-1} + a_{0,2}V_{t-1} + a_{0,3}D_{t-1} + \varepsilon_t, \quad \text{Model 1} \tag{2}$$

$$R_t = a_{0,0} + a_{0,1}R_{t-1} + a_{0,2}V_{t-1} + a_{0,3}D_{t-1} + a_{0,4}Br_{t-1} + a_{0,5}Bv_{t-1} + a_{0,6}Bd_{t-1} + \varepsilon_t, \quad \text{Model 2} \tag{3}$$

where R_t is the stock return; V_{t-1} is the stock volume; D_{t-1} is the duration between two transactions; Br_{t-1}, Bv_{t-1}, and Bd_{t-1} are Benford's variables calculated on stock returns, stock volumes, and durations respectively. ε_t is the error term, such that $\varepsilon_t \sim \text{iid}(0, \sigma^2)$.

In order to evaluate the extent to which Benford's variables contribute significant information to predict stock returns, we conducted an assessment involving two distinct models: Model 1, which excludes Benford's variables (as described in Eq. (2)); and Model 2, incorporating all factors, including Benford's variables (as outlined in Eq. (3)). However, it is imperative to emphasize that linear models require strict adherence to linearity and stationarity assumptions, as elucidated by Herrera et al. (2019), and Shobana and Umamaheswari (2021). Consequently, traditional linear models confront difficulties when attempting to proficiently forecast price movements within the intricate and unpredictable dynamics of financial markets, which are inherently marked by complexity and non-linearity (Anand, 2021). In light of these considerations, we now shift our focus to the exploration of non-linear models.

3.5.2. Smooth transition autoregressive model (STAR)

In this study, we adopt the Smooth Transition Autoregressive (STAR) model, originally developed by Luukkonen et al. (1988), Terasvirta and Anderson (1992), Teräsvirta (1994). This model serves to effectively capture market regime-switching dynamics. To further enhance our analysis, we extend the STAR model by incorporating exogenous variables, resulting in what is known as the Smooth Transition Autoregressive model with exogenous variables (STAR-X) (Dahmene et al., 2021). Our investigation encompasses the assessment of our two first-order STARX models with two distinct regimes, as delineated in Eqs. (4) and (5).

$$\begin{gathered} R_t = a_{0,0} + a_{0,1} R_{t-1} + a_{0,2} V_{t-1} + a_{0,3} D_{t-1} \\ +(a_{1,0} + a_{1,1} R_{t-1} + a_{1,2} V_{t-1} + a_{1,3} D_{t-1}) F(x_{t-d}, c, \gamma) + \varepsilon_t, \quad \text{Model 3} \end{gathered} \tag{4}$$

$$\begin{aligned} R_t = {} & a_{0,0} + a_{0,1} R_{t-1} + a_{0,2} V_{t-1} + a_{0,3} D_{t-1} + a_{0,4} Br_{t-1} + a_{0,5} Bv_{t-1} + a_{0,6} Bd_{t-1} \\ & + (a_{1,0} + a_{1,1} R_{t-1} + a_{1,2} V_{t-1} + a_{1,3} D_{t-1} + a_{1,4} Br_{t-1} + a_{1,5} Bv_{t-1} + a_{1,6} Bd_{t-1}) F(x_{t-d}, c, \gamma) \\ & + \varepsilon_t, \quad \text{Model 4} \end{aligned} \tag{5}$$

Where $F(x_{t-d}, c, \gamma)$ is the transition function, with x_{t-d} being the transition variable, d is the delay parameter, γ is the slope parameter which indicates the speed of transition between regimes, and c is the threshold parameter. ϵ_t is the error term, such that $\epsilon_t \sim$ iid $(0, \sigma^2)$. The logistic transition function is given by

$$F(x_{t-d}, c, \gamma) = \left[1 + e^{-\gamma(x_{t-d} - c)}\right]^{-1}, \gamma > 0 \tag{6}$$

An alternative to the LSTAR transition function is the exponential function (ESTAR) and it can be represented in Eq. (7):

$$F(x_{t-d}, c, \gamma) = \left[1 - e^{-\gamma(x_{t-d} - c)}\right]^{2}, \gamma > 0 \tag{7}$$

4. Results

4.1. Descriptive statistics

We begin our analysis with the presentation of summary statistics in Table 3, encompassing six time series associated with the four companies that we have randomly selected to present their estimated results in Section 4.3. Notably, all mean values for returns closely approach zero, signifying small average returns for each asset. This outcome aligns with expectations when working with high-frequency data, where short-term fluctuations dominate. Skewness statistics, serving to quantify the degree of asymmetry within the return distributions, yield noteworthy insights. In particular, skewness values for both returns and volume series predominantly exhibit negative skewness, indicating a distribution with a shorter right tail than a left tail. This observation is a common characteristic in financial data analysis, highlighting the prevalence of negative returns. Turning our attention to sample excess kurtosis values, we observe that elevated kurtosis values are indicative of distributions with fatter tails. This suggests a heightened likelihood of extreme values or outliers in the data. Furthermore, the Jarque–Bera test emphatically rejects the assumption of normality for all series. In addition, based on the outcomes of the Augmented Dickey–Fuller (ADF) test, with all p-values falling below the conventional significance level of 0.05, we establish that all series are stationary. Moreover, the correlation matrix (Appendix A, Table A.1) indicates that stock returns, stock volumes, durations, and their corresponding Benford variables exhibit weak correlations. As such, there is no substantial evidence to support the existence of a significant multicollinearity issue.

Table 3
Descriptive statistics.

	Observations	Mean	Median	Max	Min	Std. Dev.	Skewness	Kurtosis	JB P-values	ADF P-values
R_{AB}	9945	0.000	−0.001	0.085	−0.102	0.004	−0.177	119.9	0.000	0.000
V_{AB}	9945	−0.086	−0.030	6.850	−7.869	2.008	−0.026	3.480	0.000	0.000
D_{AB}	9945	487.1	11.57	228,322	0.000	7193	27.00	820.7	0.000	0.000
Br_{AB}	9945	0.176	0.152	1.657	−1.896	0.656	0.040	2.786	0.000	0.007
Bv_{AB}	9945	−2.847	−2.775	−1.701	−4.847	0.482	−0.418	2.667	0.000	0.000
Bd_{AB}	9945	−3.115	−3.069	−1.803	−5.302	0.487	−0.505	3.266	0.000	0.000
R_{ALATA}	10,247	0.000	−0.001	0.057	−0.063	0.005	−0.112	11.58	0.000	0.000
V_{ALATA}	10,247	−0.141	−0.037	11.543	−10.23	2.953	−0.076	3.829	0.000	0.000
D_{ALATA}	10,247	729.3	45.84	228,629	0.000	8793	22.02	546.3	0.000	0.000
Br_{ALATA}	10,247	−1.006	−0.947	0.548	−3.986	0.856	−0.417	3.157	0.000	0.000
Bv_{ALATA}	10,247	−3.101	−3.040	−1.919	−5.226	0.538	−0.672	3.427	0.000	0.000
Bd_{ALATA}	10,247	−3.102	−3.047	−1.928	−5.364	0.512	−0.591	3.271	0.000	0.000
R_{FORSE}	10,000	0.000	−0.001	0.065	−0.076	0.011	0.128	6.505	0.000	0.000
V_{FORSE}	10,000	0.024	0.000	7.526	−9.360	1.984	0.016	3.504	0.000	0.000
D_{FORSE}	10,000	2282	88.01	402,190	−7178	16,691	12.41	177.8	0.000	0.000
Br_{FORSE}	10,000	−1.063	−1.035	−0.015	−3.123	0.558	−0.506	2.795	0.000	0.000
Bv_{FORSE}	10,000	−2.703	−2.660	−1.611	−5.158	0.504	−0.596	3.815	0.000	0.000
Bd_{FORSE}	10,000	−0.961	−0.939	1.440	−4.015	0.691	−0.450	4.667	0.000	0.000
R_{AXA}	10,542	0.000	0.000	0.008	−0.011	0.000	−2.909	321.8	0.000	0.000
V_{AXA}	10,542	0.196	0.154	6.620	−7.998	1.818	−0.126	4.007	0.000	0.000
D_{AXA}	10,542	30.16	3.988	55,502	−3603	1113	49.66	2473	0.000	0.000
Br_{AXA}	10,542	0.581	0.628	0.828	−0.158	0.182	−1.664	5.649	0.000	0.000
Bv_{AXA}	10,542	−2.912	−2.871	−1.600	−5.919	0.579	−0.439	3.371	0.000	0.000
Bd_{AXA}	10,542	−2.706	−2.667	−0.762	−4.982	0.664	−0.189	3.206	0.000	0.000

Note: Descriptive statistics are calculated for one-second stock returns, the trading volumes, the durations, and Benford's series. JB and ADF denote the *p*-value of Jarque–Bera normality and Augmented Dickey–Fuller tests, respectively.

4.2. Benford's law first digit tests

The next step involves evaluating whether the examined time series conform to Benford's expected first-digit distribution. To achieve this, we applied three conformity diagnostics, namely the chi-square goodness-of-fit test, the Mean Absolute Deviation (MAD) measure, and the Z-statistics test, across the entire sample. The outcomes of this examination, summarized in Table 4, show that the p-values associated with the chi-square statistics for all assets fall below the 5% significance threshold, indicating that the first-digit distributions of returns, trading volumes, and durations significantly deviate from the expected Benford distribution. In addition, the return series exhibit MAD p-values exceeding the 0.015 threshold proposed by Nigrini (2020), further supporting non-conformity with Benford's Law. Conversely, the MAD p-values for the trading-volume series of ten stocks are below the 0.015 threshold, and all duration series also fall beneath this cutoff, suggesting comparatively small average deviations, although such low MAD values may still hide subtle but systematic anomalies. Consistent with these findings, the Z-statistics exceed the critical value of 1.96 (Sadaf, 2016; Alali and Romero, 2013) for all return and duration series, leading to rejection of the null hypothesis of conformity. Only seven volume series satisfy the Z-test criterion. Although no individual digit within these seven series exhibits a statistically significant deviation when evaluated in isolation, the aggregate pattern revealed by the chi-square and MAD statistics indicates substantial overall divergence from Benford's Law. Overall, such departures may signal the presence of anomalous dynamics in financial markets, potentially arising from unexpected macroeconomic shocks or firm-specific events such as regulatory changes, mergers, or dividend announcements (Nigrini, 2015).

Table 4
Chi-square Statistics test, MAD test, and Z-statistics test results.

	Stock returns				Stock volumes				Duration			
Stocks	χ^2	P-value	MAD P-value	Z-Score	χ^2	P-value	MAD	Z-Score	χ^2	P-value	MAD	Z-stat
BLV	191.89	0.0000	0.0176	−49.716	226.72	0.0000	0.0213	0.5605	31.90	0.0001	0.0048	−15.2438
ALTA	420.39	0.0000	0.0291	−46.1113	22.81	0.0036	0.0058	−6.6692	62.57	0.0000	0.0072	−6.1371
POXEL	751.68	0.0000	0.0378	−45.9053	69.87	0.0000	0.0129	−3.3740	46.60	0.0000	0.0060	−5.8916
AB	862.80	0.0000	0.0302	−45.8255	88.71	0.0000	0.0139	−2.0563	38.45	0.0000	0.0059	−9.4304
FORSE	981.06	0.0000	0.0316	−44.9574	283.00	0.0000	0.0166	−2.1557	38.35	0.0000	0.0054	−5.4484
GNFT	2202.89	0.0000	0.0485	−47.8755	57.41	0.0000	0.0109	−5.0287	17.98	0.0214	0.0041	−13.7428
SLBDS	2261.60	0.0000	0.0469	−46.3488	444.13	0.0000	0.0192	−0.8808	30.52	0.0001	0.0043	−8.4830
NACON	2319.47	0.0000	0.0708	−47.7042	118.00	0.0000	0.0152	−2.7709	44.87	0.0000	0.0060	−8.8236
WAVE	2741.30	0.0000	0.0554	−48.8241	219.02	0.0000	0.0205	−0.4063	68.47	0.0000	0.0074	−13.532
ALATA	3834.67	0.0000	0.0882	−46.1019	436.43	0.0000	0.0315	2.5656	63.83	0.0000	0.0065	−9.6284
VLTSA	3876.33	0.0000	0.0880	−48.899	142.42	0.0000	0.0168	−1.2868	27.17	0.0006	0.0037	−9.1189
ERF	8960.75	0.0000	0.0998	−45.4575	480.95	0.0000	0.0229	5.4104	43.37	0.0000	0.0056	−20.2662
ACA	13,242.49	0.0000	0.1174	−46.4602	147.51	0.0000	0.0113	−1.4323	50.12	0.0000	0.0068	−13.7076
SAF	13,464.34	0.0000	0.1174	−46.981	244.71	0.0000	0.0145	−1.9258	70.94	0.0000	0.0080	−13.531
OR	15,093.25	0.0000	0.1005	−54.568	1355.46	0.0000	0.0299	6.9715	46.71	0.0000	0.0059	−16.1776
TTE	17,069.01	0.0000	0.1263	−48.2384	128.45	0.0000	0.0094	−5.5545	20.98	0.0071	0.0038	−20.759
AF	17,087.37	0.0000	0.1173	−48.1893	133.16	0.0000	0.0115	−3.8245	24.50	0.0018	0.0045	−18.5025
AXA	18,964.31	0.0000	0.1353	−47.835	98.85	0.0000	0.0103	−1.4280	38.12	0.0000	0.0061	−6.1298
ALO	34,743.36	0.0000	0.1298	−49.785	182.27	0.0000	0.0134	−1.7306	38.76	0.0000	0.0056	−12.0339
NEOEN	36,609.79	0.0000	0.1280	−48.5632	217.32	0.0000	0.0145	−2.4560	28.62	0.0037	0.0053	−9.7673

Note: A Z-statistic with an absolute value of 1.96 corresponds to a p-value of 0.05 under the standard normal distribution. Consequently, when the computed Z-statistic is 1.96 or higher , the null hypothesis is rejected at the 5% significance level.

4.3. Estimation results

Before presenting the results of the linear and nonlinear models for the assets under investigation, we apply an 80%/20% cross-validation method to compare the two models using the Root Mean Squared Error (RMSE). Models 1 and 3 exclude Benford's variables, while Models 2 and 4 include them. Following this, we proceed to estimate our models. Initially, we regress all variables from the two models as specified in Eqs. (2) and (3) for the linear regression, and Eqs. (4) and (5) for the smooth transition autoregressive models. We then apply a backward elimination method to remove non-significant explanatory variables, starting with those having the highest P-values. This process continues until removing an explanatory variable no longer enhances the model's performance based on AIC and BIC criteria. In the next two subsections, we present the RMSE calculated on the training and testing sets for all assets. Subsequently, we run our model estimations on the entire dataset. For companies where Benford variables do seem significant, we compare the AIC and BIC criteria (and the HQ criteria for the STARX models). Finally, we present the estimated results for the four randomly selected companies.

4.3.1. Estimates of linear models

Among the twenty assets under investigation, Benford's variables were significant in only 25% of the cases in the linear regression. As shown in Table 5, the RMSE for the training and test sets in Model 2 are similar to those in Model 1. Consequently, according to the AIC criteria (see Appendix B, Table B.1), Model 2, which includes Benford's variables, provides a better fit for our data. Estimation results for the first-order linear models specified in Eq. (2), and Eq. (3) for AB and ALATA assets are presented in Table 6. The table reveals that Benford's variable, computed from stock returns (Br_{t-1}), has a positive and statistically significant effect on stock returns for AB assets, while it has a negative and statistically significant effect for ALATA assets in the second model. Additionally, Benford's variable calculated from trading volumes (Bv_{t-1} shows a negative and statistically significant impact on stock returns for AB assets. Consequently, we proceed to investigate the potential existence of a nonlinear relationship between these variables.

Table 5
Evaluation of RMSE metric for the linear models.

	Linear model with Benford's variables		Linear model without Benford's variables	
	$RMSE_{Train}$	$RMSE_{Test}$	$RMSE_{Train}$	$RMSE_{Test}$
AB	0.00339	0.00402	0.00339	0.00402
ALATA	0.00501	0.00507	0.00501	0.00507
AXA	0.00028	0.00022	0.00028	0.00022
TTE	0.00015	0.00013	0.00015	0.00013
SLBDS	0.00421	0.00399	0.00421	0.00399

Table 6
Estimates of linear models.

	Model 1		Model 2	
	Coeff	*P*-value	Coeff	*P*-value
AB				
Const	−0.00002	0.5445	−0.00046	0.0286**
R_{t-1}	−0.14672	0.0000***	−0.14756	0.0000***
V_{t-1}	−0.00004	0.0170**	−0.00004	0.0177**
D_{t-1}	DEL		DEL	
Br_{t-1}			0.00012	0.0217**
Bv_{t-1}			−0.00016	0.0262**
Bd_{t-1}			DEL	
R^2	0.02206		0.02305	
AIC	−8.45339		−8.45400	
BIC	−8.45122		−8.45038	
ALATA				
Const	−0.00009	0.0628*	−0.00021	0.0061***
R_{t-1}	−0.26783	0.0000***	−0.26811	0.0000***
V_{t-1}	DEL		DEL	
D_{t-1}	DEL		DEL	
Br_{t-1}			−0.00011	0.0435**
Bv_{t-1}			DEL	
Bd_{t-1}			DEL	
R^2	0.0717		0.0721	
AIC	−7.7476		−7.7478	
BIC	−7.7462		−7.7457	

*** indicates statistical significance at 1 percent,
** indicates statistical significance at 5 percent,
* indicates statistical significance at 10 percent.
Note: DEL: DELATED.

4.3.2. Estimates of nonlinear models

To identify and estimate STAR (Smooth Transition Autoregressive) models, we adhere to the comprehensive four-step methodology outlined by Teräsvirta (1994). The first step entails selecting the appropriate lag length, a critical aspect for identifying and estimating linear autoregressive models within each regime. In our study, we determined the autoregressive lag in each regime based on parameter significance, while imposing $d = 1$ for the transition variable to maintain consistency with a single autoregressive lag. This choice was also substantiated by formal tests, as described in Teräsvirta (1994). The second step of our methodology involves assessing linearity and identifying the appropriate transition variable for the STARX model. Following the established procedure of Luukkonen et al. (1988) and the selection rule described in Tsay (1989), we evaluate several potential transition variables, R_{t-d}, V_{t-d}, D_{t-d}, Br_{t-d}, Bv_{t-d}, and Bd_{t-d}, using the LM linearity test based on the third-order Taylor series approximation of the STAR specification. Under the null hypothesis of linearity, rejection of the LM test indicates that the candidate variable induces statistically significant nonlinearity in the conditional mean and therefore qualifies as a valid transition variable. The results of the linearity tests are provided in Appendix C, Table C.1, and are reported for eleven assets under study. When multiple candidate variables reject linearity, we follow standard practice and select the variable with the strongest rejection, i.e., the variable with the lowest *p*-value, as the transition variable. This data-driven criterion is also used to determine whether a logistic STAR specification (LSTAR) or an exponential specification (ESTAR) is appropriate (results documented in Appendix C, Table C.2). For instance, Table C.1 in Appendix C shows that, for the FORSE asset, the null hypothesis of linearity is rejected when the transition variables are R_{t-1} and D_{t-1}, and similarly for the AXA asset when considering V_{t-1} and D_{t-1} in Model 3 (Eq. (4)). In both cases, D_{t-1} displays the lowest marginal probability value and is therefore selected as the transition variable. The results in Model 4 (Eq. (5)) show that R_{t-1} and Br_{t-1} are the most suitable transition variables for FORSE and AXA assets, respectively, following the same criterion. The rejection of the null hypothesis of linearity hints at the possibility of a nonlinear relationship between one-second stock returns and Benford's variables. Subsequently, the final step focuses on estimating the parameter values for the STARX models on the entire dataset. The estimation process employs nonlinear least squares.

Out of the twenty assets examined, Benford's variables proved significant in only 55% of the cases within the nonlinear regression analysis. As demonstrated in Table 7, the RMSE for both the training and test sets is lower in Model 4 compared to Model 3. This suggests that Model 4, which includes Benford's variables, should be retained for predicting intraday stock returns. Furthermore, based on the AIC, BIC, and HQ (the Hannan–Quinn information criterion) criteria (see Appendix B, Table B.2), Model 2 offers a better fit for our data. Table 8 provides an overview of the estimated results for the final STARX models for FORSE and AXA assets, summarizing the key findings of our analysis. Notably, for FORSE assets, the Model 4 results indicate that when R_{t-1} is the transition variable, stock returns are significant in both regimes, with a negative sign in the lower regime and a positive sign in the upper regime. Additionally, we observe that Benford's variables calculated from trading volumes and durations appear to be significant at the 5 percent level in the upper regime, with negative and positive signs, respectively.

Table 7
Evaluation of RMSE metric for the STARX models.

	STARX model with Benford's variables		STARX model without Benford's variables	
	$RMSE_{Train}$	$RMSE_{Test}$	$RMSE_{Train}$	$RMSE_{Test}$
AB	0.00335	0.00393	0.00336	0.00401
ALATA	0.00497	0.00501	0.00498	0.00504
BLV	0.00305	0.00631	0.00306	0.00633
POXEL	0.00544	0.00669	0.00545	0.00668
VLTSA	0.00273	0.00243	0.00272	0.00254
AF	0.00061	0.00082	0.00061	0.00083
ALO	0.00045	0.00036	0.00044	0.00036
AXA	0.00027	0.00023	0.00029	0.00023
FORSE	0.00986	0.01080	0.00987	0.01080
SLBDS	0.00417	0.00394	0.00420	0.00396
WAVE	0.00356	0.00254	0.00355	0.00255

Table 8
Estimates of two regime STAR-X models for FORSE and AXA assets.

		Model 3		Model 4	
		Coeff	P-value	Coeff	P-value
FORSE					
Transition variable		D_{t-1}		R_{t-1}	
Regime 1	Const	−0.00157	0.0003***	−0.00079	0.1542
	R_{t-1}	−0.17981	0.0000***	−0.36398	0.0000***
	V_{t-1}	0.00011	0.5096	DEL	
	D_{t-1}	0.00000	0.4354	0.00000	0.0464**
	Br_{t-1}			DEL	
	Bv_{t-1}			−0.00027	0.1807
	Bd_{t-1}			0.00004	0.7795
Regime 2	Const	0.00645	0.0867*	0.00808	0.2342
	R_{t-1}	−0.88191	0.0462**	0.34100	0.0000***
	V_{t-1}	−0.00062	0.3096	DEL	
	D_{t-1}	0.00000	0.8279	−0.00000	0.0723*
	Br_{t-1}			DEL	
	Bv_{t-1}			−0.00295	0.0273**
	Bd_{t-1}			0.00750	0.0001***
	γ	0.44099	0.1463	1.43124	0.0819*
	c	−2926.30	0.4041	−0.02973	0.0000***
Metrics	AIC	−9.1951		−9.1953	
	BIC	−9.1871		−9.1859	
	HQ	−9.1924		−9.1921	
AXA					
Transition variable		D_{t-1}		Br_{t-1}	
Regime 1	Const	0.00201	0.0469**	−0.00004	0.2948
	R_{t-1}	−3.42307	0.0129**	−0.09045	0.0000***
	V_{t-1}	0.00653	0.0000***	0.00000	0.5108
	D_{t-1}	0.00001	0.0000***	0.00000	0.0000***
	Br_{t-1}			−0.00010	0.0001***
	Bv_{t-1}			−0.00002	0.0054***
	Bd_{t-1}			0.00000	0.7529
Regime 2	Const	−0.00209	0.0449**	0.00273	0.0110**
	R_{t-1}	3.45458	0.0128**	0.06387	0.6557
	V_{t-1}	−0.00681	0.0000***	−0.00002	0.2812
	D_{t-1}	−0.00001	0.0000***	−0.00000	0.1987
	Br_{t-1}			−0.00295	0.0194**
	Bv_{t-1}			0.00011	0.2484
	Bd_{t-1}			−0.00001	0.8936
	γ	4.34943	0.0008***	1.43124	0.0001***
	c	−780.191	0.0046***	0.91438	0.0000***
Metrics	AIC	−16.4549		−16.4560	
	BIC	−16.4455		−16.4450	
	HQ	−16.4539		−16.4530	

Note: ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels. DEL = delayed.

This suggests that deviations from Benford in the volume and duration data can signal fraudulent events influencing stock returns in the upper regime. For AXA assets, a striking observation is made in Model 4. When the transition variable is set as Br_{t-1}, the first lag of stock returns, the duration, Benford's variable calculated from stock returns, and Benford's variable calculated from trading volumes exhibit significance at the 1 percent level in the lower regime. In contrast, in the upper regime, only Br_{t-1} shows significance with a negative sign. This finding can be attributed to the ability of the Benford variable to detect abnormal regimes when stock returns deviate significantly from a threshold parameter. It is also worth noting that the estimates of the speed of transition (γ) suggest a smooth transition between the two regimes for FORSE and AXA assets. Furthermore, it is noteworthy that all threshold variables are consistently significant at the 1 percent level. To sum up, the nonlinear STARX specifications provide stronger evidence of the predictive value of Benford-based variables compared to the linear models for predicting intraday stock returns. In the linear framework, Benford's variables are significant in only 25% of the estimated cases, and the improvement in model fit (as measured by RMSE and AIC) is limited. By contrast, the nonlinear STARX models show a higher proportion of significant Benford effects, improved goodness-of-fit across information criteria, and clearer predictive gains. Moreover, the STARX estimation results also uncover pronounced regime-dependent patterns that cannot be captured by the linear models. For example, in the FORSE asset, key variables, including Benford measures, are significant only within specific regimes, and their signs change across regimes. Similarly, in the AXA asset, several Benford-based variables are highly significant in the lower regime when the transition variable is Br_{t-1}, whereas only Br_{t-1} remains significant in the upper regime.

These asymmetric effects are not detectable in a linear specification and indicate that deviations from Benford's law signal abnormal market conditions that become relevant only when the system crosses certain thresholds. The significance of the threshold variables and the smooth estimated transition parameters further confirms the presence of nonlinear dynamics. Overall, the nonlinear specifications better account for the underlying dynamics of the data and offer more informative insights into when and how Benford deviations matter for stock return behavior.

4.4. Post-estimate examination

4.4.1. Transition functions

After analyzing the coefficients within the estimated STAR-X models, it becomes imperative to delve into the examination of the resulting transition functions. Fig. 1 provides a visual representation of the estimated transition functions over time and transition variables of the retained models for the FORSE and AXA assets. We start with FORSE assets, where the estimated threshold value of R_{t-1} registers at −0.029, accompanied by a slope parameter γ of 1.43. This low γ value highlights a smooth transition of F from 0 to 1. For the AXA asset, the estimated threshold value of Br_{t-1} stands at 0.91, and the slope parameter γ is notably low at 1.43. This model demonstrates a smooth shift between the two regimes, primarily due to the slow rate of transition. Notably, the majority of observations reside within the regime where $F(Br_{t-1})$ is less than 0.12. Additionally, the scatter plot visually confirms that the estimated model distinctly aligns with an LSTAR model, with most observations clustered in the left-hand tail of the logistic function.

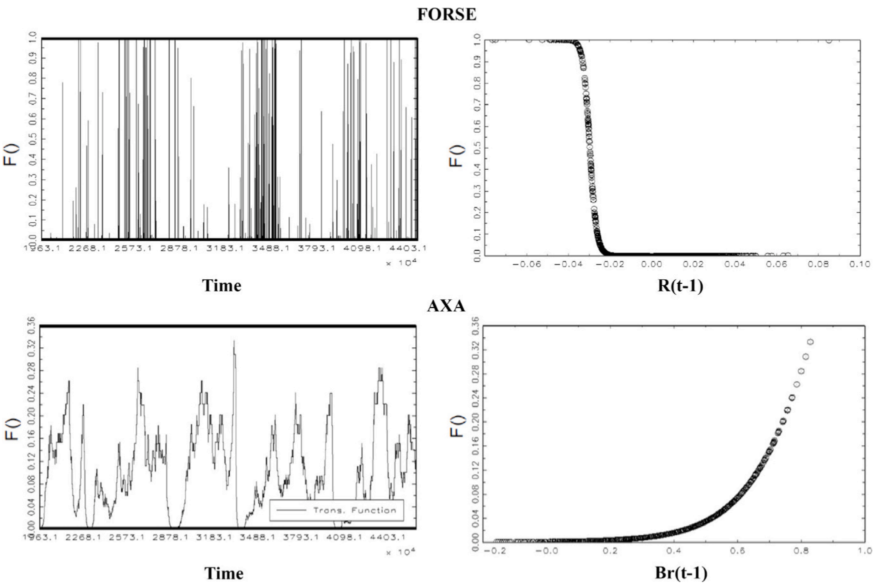

Fig. 1. Transition functions in the estimated STAR-X models for FORSE and AXA assets. *Note:* The figure shows the transition functions over time (left panels) and the scatter plots of $F(\cdot)$ against R_{t-1} for FORSE and Br_{t-1} for AXA (right panels).

4.4.2. Impulse response functions

To better illustrate the dynamic behavior of our STAR-X estimated models and gain insight into how R responds to external shocks, we delve into the persistence of these shocks using a non-linear impulse response function. Impulse response functions in non-linear models exhibit historical dependencies, with their characteristics influenced by both the direction and magnitude of the shocks. In Section 3.5.2, our analysis of the estimated results for our chosen models (Model 4) reveals that the threshold variable and value are as follows: R_{t-1} and −0.029 for FORSE assets, and Br_{t-1} and 0.91 for AXA assets. To streamline our analysis and concentrate on the non-linear effects of Benford's variables on one-second stock returns predictions, we divide the entire dataset into two regimes based on the threshold value. Regime 1 comprises data where the threshold variable is less than or equal to the threshold value, while Regime 2 encompasses data where the threshold variable exceeds the threshold value. Fig. 2 illustrates the impulse responses within the retained STARX models of FORSE and AXA assets, which exhibit two distinct regimes. For FORSE assets, the responses of R to its shocks exhibit convergence patterns in the lower regime and asymmetry in the two regimes, indicating that the relationship between stock returns and shocks changes smoothly between different states of conditions. Moving on to showcase the impulse responses within the estimated STARX model for AXA assets, we observe that the reactions of variable R to both positive and negative R shocks display subtle asymmetries in both regimes. These findings align with the estimated coefficients of the retained Model 4, as detailed in Table 8.

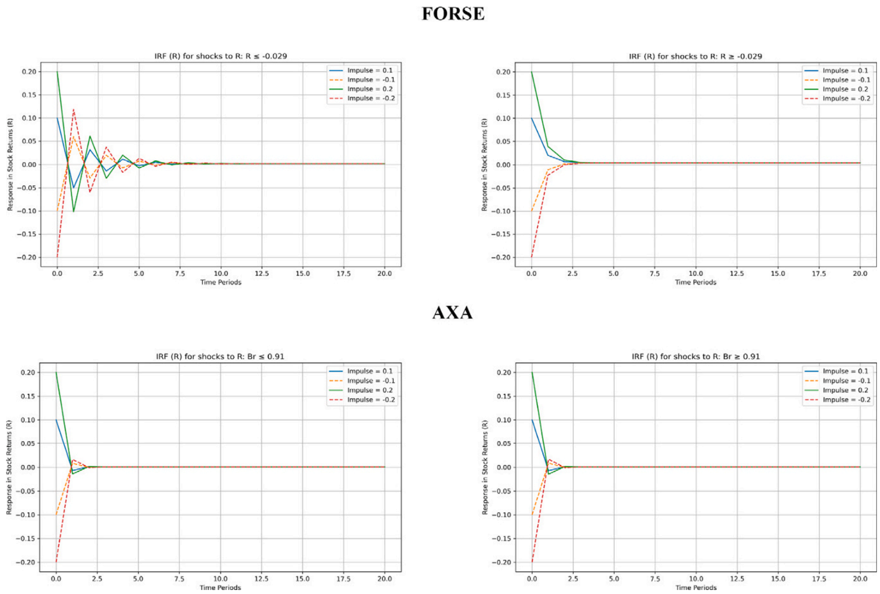

Fig. 2. Impulse response function in two regime STARX model for FORSE and AXA assets. *Note:* The figure depicts impulse responses from the retained STAR-X models for the FORSE and AXA assets, distinguishing between two regimes. The left panels correspond to the regime in which the transition variable is below the threshold, and the right panels correspond to the regime in which the transition variable exceeds the threshold.

4.5. Discussion

Benford's Law is widely used to detect irregularities in financial data, yet it does not necessarily indicate market inefficiencies, an inherent characteristic of financial systems. Instead, it highlights anomalies that may arise from various factors, including liquidity constraints, transaction costs, and trading strategies. In this study, we analyze intraday stock returns, trading volumes, and duration series, revealing significant deviations from Benford's distribution. While such deviations are often linked to fraudulent activities, they can also stem from non-fraudulent causes, such as liquidity issues or order-based manipulation strategies. To further explore these anomalies, we examined major corporate events affecting the companies studied between January 3 and December 31, 2022. For instance, in a decision issued on 24 March and made public on 28 March 2022, the AMF's sanctions committee imposed a fine on AB Science company for failing to promptly disclose privileged information about the marketing authorization dossier for masitinib, intended for the treatment of mastocytosis. Additionally, the Atari Group posted a current operating profit of −2.8 ME in the first half of 2021–2022, affected by the effects of the COVID-19 crisis. Furthermore, on February 17, 2022, Forsee Power published that it exceeded its 2021 turnover target with revenues up 17% to 72.4 million €. Moreover, AXA announced the finalization of a 691 million € sale of its subsidiary AXA Banque Belgique to the Belgian cooperative bank Crelan on December 31, 2021. These significant corporate events, while unrelated to fraud, may contribute to deviations from Benford's Law. Previous research supports this view, showing that deviations from Benford's distribution may yield false positives, as seen in cryptocurrency markets (Karavardar, 2014; Vičič and Tošić, 2022). Beyond corporate events, market microstructure factors can also explain these anomalies. Dalko and Wang (2020) highlighted order-based manipulation (OBM) as a key driver of high-frequency data distortions, where traders submit and cancel large orders to create an illusion of liquidity. This technique, which accounted for 99.5% of order cancellations in CAC 40 stocks (Anagnostidis and Fontaine, 2020), is a well-documented market manipulation strategy. Additionally, Erdemlioglu et al. (2021) demonstrated that intraday price jumps are influenced not only by news events but also by technical trading strategies, particularly in smaller firms and energy markets. Liquidity constraints further contribute to deviations from Benford's Law. Transaction costs, limited market depth, and short-selling constraints can create price distortions, particularly in low-liquidity assets. In this study, we capture liquidity effects through transaction volume data and measure their divergence from Benford's expected distribution. These insights suggest that Benford's Law could serve as a tool to quantify market illiquidity, as deviations from the expected distribution may signal liquidity shortages rather than fraudulent activity. Overall, the aim of this study is not to provide a predictive model for intraday stock returns based on Benford's variables with a large degree of accuracy, because of the efficient behavior of financial markets. Most of the time, companies operate in an efficient regime where Benford's variables are not significant, and other factors play a more crucial role. However, during periods of market anomalies, whether driven by intentional manipulation or unforeseen irregularities, Benford's variables become more informative, offering early signals of abnormal activity. Our findings suggest that these deviations, when identified in returns, trading volumes, and trade durations, have condition-specific effects and can precede significant price movements, presenting potential trading opportunities. Importantly, the fact that Benford-derived variables become statistically significant only in a subset of cases does not indicate limited robustness. Rather, it reflects their inherent conditional relevance. Moreover, we find that the statistical significance of Benford's metrics correlates with the predictive strength of our models, regardless of whether the variables are explicitly included. Crucially, nonlinear frameworks, particularly smooth transition models, demonstrate superior performance in capturing these irregular patterns compared to linear models. These results reinforce the practical relevance of Benford's Law as a complementary tool for detecting market irregularities and enhancing short-term forecasting strategies in high-frequency trading environments.

4.6. Limitations and real-world applications for finance professionals

As previously discussed, there are certain limitations to applying Benford's Law to financial data. However, due to our specific methodology, our study remains unaffected by these constraints. We addressed the sample size sensitivity by using moving windows of 200 consecutive transactions and normalizing the chi-squared test statistics by the sample size N, thus mitigating the impact of larger samples leading to higher deviations despite unchanged relative (Leemis et al., 2000). Nevertheless, we acknowledge the limitation of relying solely on the chi-squared statistics. Future studies could enhance robustness by incorporating alternative tests such as the mean absolute deviation and the Z-statistics. Furthermore, following (Sifat et al., 2024), the applicability of Benford's Law to numbers with at least three digits remains a debated issue that deserves an accurate analysis in future research. Expanding this research to different market contexts would also provide valuable insights for future investigations.

Our findings also carry several important economic implications. The significance of Benford-based variables, particularly within nonlinear regimes, suggests that deviations from expected digit distributions contain information about underlying market conditions. Such deviations may arise when traders behave abnormally, when liquidity becomes constrained, or when market participants engage in coordinated or manipulative trading practices. In this sense, Benford's indicators can serve as early warning signals of market stress or irregular activity, complementing traditional technical measures such as Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), or volatility-based indicators. The regime-dependent effects observed in the nonlinear models further imply that the informational content of Benford deviations intensifies during periods of transition or instability, when markets move away from normal trading patterns. These insights suggest that incorporating Benford-based diagnostics into monitoring systems could help analysts better detect market inefficiencies, anticipate abrupt shifts in asset dynamics, and improve risk management during episodes of heightened uncertainty. In addition to its relevance for market participants, Benford's Law also holds considerable potential for authorities and legal institutions. Importantly, U.S. courts have deemed analyses based on Benford's Law as legally admissible, which underscores the methodological credibility of this approach in formal investigations. This recognition creates an avenue for agencies such as the Securities and Exchange Commission (SEC) and the Commodity Futures Trading Commission (CFTC), as well as European authorities like the European Securities and Markets Authority (ESMA), the Autorité des Marches Financiers (AMF), to integrate Benford-based diagnostics into their monitoring procedures. Embedding these tools within regulatory systems would allow institutions to leverage a quantitative, data-driven method for identifying irregular trading patterns or early signs of market manipulation. By doing so, oversight bodies could enhance their capacity for preventive intervention, strengthen enforcement actions, and improve the detection of fraudulent behavior before it escalates into broader financial misconduct.

5. Conclusion

The ongoing challenge of forecasting high-frequency stock returns has prompted the exploration of unconventional statistical tools. This study investigates the predictive relevance of Benford's Law in intraday market dynamics by analyzing the distribution of first digits in stock returns, trading volumes, and trade durations for twenty companies listed on Euronext Paris. Significant deviations from the expected Benford distribution, detected using chi-squared analysis, the MAD test, and the Z-statistic, suggest the presence of market irregularities that may not be apparent through traditional indicators. By incorporating Benford-based variables into both linear and Smooth Transition Autoregressive (STAR) models, we find that their predictive power is regime-dependent, gaining particular relevance during periods characterized by abnormal trading behavior. These findings suggest that Benford's Law, while not a direct measure of inefficiency or fraud, can serve as a valuable diagnostic tool for identifying shifts in market behavior at a granular level. Importantly, deviations from Benford's distribution should not be viewed solely as evidence of manipulation (Karavardar, 2014), but rather as signals of structural irregularities that may stem from a range of market phenomena, including algorithmic trading, liquidity shocks, or order flow imbalances. When used in conjunction with conventional technical indicators, Benford-based metrics can enhance the robustness and responsiveness of intraday trading strategies. This research contributes to the growing literature on statistical forensics in finance and offers practical insights for traders, analysts, and regulatory bodies. Future work could extend this approach to broader markets and asset classes or explore integration with machine learning frameworks to improve the detection and forecasting of short-term anomalies in real-time trading environments.

CRediT authorship contribution statement

Amal Ben Hamida: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Christian de Peretti:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Methodology, Conceptualization. **Lotfi Belkacem:** Visualization, Validation, Supervision, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Appendix A. Correlation matrix and the backward method procedure

A.1. Correlation matrix

See Table A.1.

Table A.1
Correlation matrix of the four presented stock assets.

	R_{AB}	V_{AB}	D_{AB}	Br_{AB}	Bv_{AB}	Bd_{AB}	R_{ALATA}	V_{ALATA}	D_{ALATA}	Br_{ALATA}	Bv_{ALATA}	Bd_{ALATA}	R_{FORSE}	V_{FORSE}	D_{FORSE}	Br_{FORSE}	Bv_{FORSE}	Bd_{FORSE}	R_{AXA}	V_{AXA}	D_{AXA}	Br_{AXA}	Bv_{AXA}	Bd_{AXA}
R_{AB}	1	−0.023	0.010	−0.020	−0.019	0.010	0.008	0.003	0.009	0.017	−0.013	−0.006	0.002	−0.008	0.006	0.001	−0.009	−0.006	−0.005	0.002	0.005	0.011	0.008	−0.004
V_{AB}	−0.023	1	0.043	0.001	0.006	0.010	0.015	0.032	−0.011	−0.012	−0.003	0.014	−0.004	−0.002	0.018	0.010	−0.009	0.000	0.005	0.010	0.004	0.014	−0.009	0.009
D_{AB}	0.010	0.043	1	0.022	−0.004	0.012	0.004	−0.014	−0.003	−0.002	0.004	−0.011	−0.001	−0.008	−0.004	−0.001	−0.011	−0.002	0.004	−0.005	−0.001	−0.001	−0.013	0.010
Br_{AB}	−0.020	0.001	0.022	1	−0.019	0.053	0.006	−0.013	−0.001	−0.170	0.049	0.112	0.002	0.000	−0.015	−0.279	−0.036	−0.299	0.004	0.013	0.008	−0.122	0.092	−0.123
Bv_{AB}	−0.019	0.006	−0.004	−0.019	1	0.099	−0.001	−0.008	−0.010	−0.177	−0.034	−0.003	0.003	0.001	−0.008	0.034	0.001	0.198	−0.012	0.004	−0.007	−0.022	−0.107	0.089
Bd_{AB}	0.010	0.010	0.012	0.053	0.099	1	0.002	0.007	0.003	0.008	0.016	−0.095	−0.004	−0.005	−0.007	0.092	0.106	−0.017	0.007	−0.009	0.014	−0.026	0.074	0.058
R_{ALATA}	0.008	0.015	0.004	0.006	−0.001	0.002	1	0.027	−0.005	−0.016	−0.001	0.006	−0.016	−0.006	−0.012	−0.007	0.010	0.002	−0.003	0.001	0.000	−0.006	0.008	−0.008
V_{ALATA}	0.003	0.032	−0.014	−0.013	−0.008	0.007	0.027	1	0.001	0.010	0.010	−0.012	−0.011	0.015	0.003	0.002	−0.018	−0.018	0.017	0.008	0.021	0.004	0.001	−0.001
D_{ALATA}	0.009	−0.011	−0.003	−0.001	−0.010	0.003	−0.005	0.001	1	0.006	−0.016	−0.027	0.024	−0.010	−0.005	−0.011	0.006	−0.014	−0.004	0.003	−0.001	−0.007	0.008	−0.005
Br_{ALATA}	0.017	−0.012	−0.002	−0.170	−0.177	0.008	−0.016	0.010	0.006	1	−0.101	−0.052	0.005	−0.009	0.003	0.076	0.014	−0.004	−0.003	0.009	−0.004	−0.024	0.118	−0.129
Bv_{ALATA}	−0.013	−0.003	0.004	0.049	−0.034	0.016	−0.001	0.010	−0.016	−0.101	1	−0.034	−0.011	0.002	0.007	0.045	0.009	−0.186	−0.002	0.002	0.002	0.225	0.056	−0.028
Bd_{ALATA}	−0.006	0.014	−0.011	0.112	−0.003	−0.095	0.006	−0.012	−0.027	−0.052	−0.034	1	0.002	−0.002	0.005	0.015	−0.036	−0.099	−0.006	0.016	−0.015	0.101	−0.024	0.080
R_{FORSE}	0.002	−0.004	−0.001	0.002	0.003	−0.004	−0.016	−0.011	0.024	0.005	−0.011	0.002	1	−0.035	0.119	−0.004	−0.010	0.004	0.000	−0.019	−0.002	−0.002	−0.011	0.001
V_{FORSE}	−0.008	−0.002	−0.008	0.000	0.001	−0.005	−0.006	0.015	−0.010	−0.009	0.002	−0.002	−0.035	1	0.020	0.014	0.003	0.001	0.008	−0.007	−0.002	−0.007	−0.016	0.011
D_{FORSE}	0.006	0.018	−0.004	−0.015	−0.008	−0.007	−0.012	0.003	−0.005	0.003	0.007	0.005	0.119	0.020	1	0.017	0.008	0.019	0.023	0.009	−0.003	0.008	0.001	−0.014
Br_{FORSE}	0.001	0.010	−0.001	−0.279	0.034	0.092	−0.007	0.002	−0.011	0.076	0.045	0.015	−0.004	0.014	0.017	1	0.036	0.173	−0.011	0.011	−0.008	0.140	−0.077	0.348
Bv_{FORSE}	−0.009	−0.009	−0.011	−0.036	0.001	0.106	0.010	−0.018	0.006	0.014	0.009	−0.036	−0.010	0.003	0.008	0.036	1	0.165	−0.004	−0.018	−0.008	−0.111	0.273	−0.005
Bd_{FORSE}	−0.006	0.000	−0.002	−0.299	0.198	−0.017	0.002	−0.018	−0.014	−0.004	−0.186	−0.099	0.004	0.001	0.019	0.173	0.165	1	−0.008	−0.014	−0.010	0.037	−0.090	−0.002
R_{AXA}	−0.005	0.005	0.004	0.004	−0.012	0.007	−0.003	0.017	−0.004	−0.003	−0.002	−0.006	0.000	0.008	0.023	−0.011	−0.004	−0.008	1	−0.042	0.016	−0.019	−0.006	−0.001
V_{AXA}	0.002	0.010	−0.005	0.013	0.004	−0.009	0.001	0.008	0.003	0.009	0.002	0.016	−0.019	−0.007	0.009	0.011	−0.018	−0.014	−0.042	1	−0.071	0.008	−0.009	0.010
D_{AXA}	0.005	0.004	−0.001	0.008	−0.007	0.014	0.000	0.021	−0.001	−0.004	0.002	−0.015	−0.002	−0.002	−0.003	−0.008	−0.008	−0.010	0.016	−0.071	1	0.015	−0.016	−0.007
Br_{AXA}	0.011	0.014	−0.001	−0.122	−0.022	−0.026	−0.006	0.004	−0.007	−0.024	0.225	0.101	−0.002	−0.007	0.008	0.140	−0.111	0.037	−0.019	0.008	0.015	1	0.039	0.181
Bv_{AXA}	0.008	−0.009	−0.013	0.092	−0.107	0.074	0.008	0.001	0.008	0.118	0.056	−0.024	−0.011	−0.016	0.001	−0.077	0.273	−0.090	−0.006	−0.009	−0.016	0.039	1	−0.049
Bd_{AXA}	−0.004	0.009	0.010	−0.123	0.089	0.058	−0.008	−0.001	−0.005	−0.129	−0.028	0.080	0.001	0.011	−0.014	0.348	−0.005	−0.002	−0.001	0.010	−0.007	0.181	−0.049	1

A.2. Backward method procedure

See Table A.2.

Table A.2
Estimates of two two-regime STAR-X complete model for FORSE assets.

FORSE (Model 4 Initial)			
Transition variable		R_{t-1}	
		Coeff	*P*-value
Regime 1	Const	−0.00089	0.1266
	R_{t-1}	−0.36434	0.0000***
	V_{t-1}	−0.00003	0.6227
	D_{t-1}	0.00000	0.0435**
	Br_{t-1}	−0.00012	0.5341
	Bv_{t-1}	−0.00027	0.1886
	Bd_{t-1}	0.00006	0.7103
Regime 2	Const	0.00420	0.5484
	R_{t-1}	0.36134	0.0000***
	V_{t-1}	0.00027	0.6141
	D_{t-1}	−0.00000	0.0588*
	Br_{t-1}	−0.00369	0.0968*
	Bv_{t-1}	−0.00490	0.0169**
	Bd_{t-1}	0.00861	0.0000***
	γ	0.68987	0.0942*
	c	−0.02931	0.0000***
Metrics	AIC	−9.1947	
	BIC	−9.1825	
	HQ	−9.1906	
FORSE (V_{t-1} delated)			
Transition variable		R_{t-1}	
Regime 1	Const	−0.00089	0.1253
	R_{t-1}	−0.36410	0.0000***
	V_{t-1}	0.00000	0.0445**
	D_{t-1}	0.00000	0.0435**
	Br_{t-1}	−0.00012	0.5290
	Bv_{t-1}	−0.00027	0.1880
	Bd_{t-1}	0.00006	0.7086
Regime 2	Const	0.00449	00.5169
	R_{t-1}	0.36067	0.0000***
	V_{t-1}	−0.00000	0.0565*
	D_{t-1}	−0.00353	0.1022
	Br_{t-1}	−0.00486	0.0173**
	Bv_{t-1}	0.00853	0.0000***
	Bd_{t-1}	0.69042	0.0000***
	γ	0.68987	0.0942*
	c	−0.02931	0.0000***
Metrics	AIC	−9.1951	
	BIC	−9.1843	
	HQ	−9.1914	

*** indicates statistical significance at 1 percent,
** indicates statistical significance at 5 percent,
* indicates statistical significance at 10 percent.
Note: DEL: DELATED.

Appendix B. Information criteria results for linear and nonlinear models

B.1. AIC and BIC criteria results for linear models

See Table B.1.

Table B.1
AIC and BIC criteria results for the linear models.

Stocks	Metrics	Linear model without Benford's variables	Linear model with Benford's variables
AB	AIC[a]	−8.4534	−8.4540
	BIC[a]	−8.4512	−8.4504
ALATA	AIC	−7.7476	−7.7478
	BIC	−7.7462	−7.7457
AXA	AIC	−13.6165	−13.6166
	BIC	−13.6139	−13.6117
TTE	AIC	−14.7965	−14.7967
	BIC	−14.7940	−14.7918
SLBDS	AIC	−8.1200	−8.1206
	BIC	−8.1171	−8.1155

[a] AIC is the Akaike information criterion and BIC is the Bayesian information criterion.

B.2. AIC, BIC, and HQ information criteria results for the STARX models

See Table B.2.

Table B.2
AIC, BIC, and HQ information criteria results for the STARX models.

Stocks	Metrics	STARX model without Benford's variables	STARX model with Benford's variables
AB	AIC	−11.3020	−11.3040
	BIC	−11.2950	−11.2920
	HQ	−11.3000	−11.3000
ALATA	AIC	−10.5960	−10.5970
	BIC	−10.5890	−10.5850
	HQ	−10.5940	−10.5940
BLV	AIC	−11.0700	−11.0700
	BIC	−11.0630	−11.0600
	HQ	−11.0680	−11.0670
POXEL	AIC	−10.3170	−10.3180
	BIC	−10.3100	−10.3060
	HQ	−10.3150	−10.3140
VLTSA	AIC	−11.8280	−11.8370
	BIC	−11.8220	−11.8280
	HQ	−11.8260	−11.8340
AF	AIC	−14.6270	−14.6300
	BIC	−14.6210	−14.6200
	HQ	−14.6250	−14.6260
ALO	AIC	−15.4500	−15.4520
	BIC	−15.4430	−15.4440
	HQ	−15.4470	−15.4480
AXA	AIC	−16.4549	−16.4560
	BIC	−16.4455	−16.4450
	HQ	−16.4539	−16.4530
FORSE	AIC	−9.1951	−9.1953
	BIC	−9.1871	−9.1859
	HQ	−9.1924	−9.1921
SLBDS	AIC	−10.9620	−10.9640
	BIC	−10.9550	−10.9550
	HQ	−10.9600	−10.9610
WAVE	AIC	−11.3760	−11.3780
	BIC	−11.3690	−11.3700
	HQ	−11.3740	−11.3750

AIC is the Akaike information criterion, BIC is the Bayesian information criterion, and HQ is the HannanQuinn Information criterion.

Appendix C. Linearity tests and transition selection procedure

C.1. Linearity tests

Following the approach outlined in Luukkonen et al. (1988), the Lagrange multiplier linearity test was conducted, employing a Taylor series approximation of the nonlinear STAR model. The objective was to identify which explanatory variables qualify as valid transition variables. Substituting the transition function with a third-order Taylor approximation within the STAR model yields the following Equation Eq. (C.1):

$$y_t = \phi_0 + \phi_1 y_{t-1} + \sum_{i=1}^{3} \phi_{1+i} y_{t-1} x_t^i + \epsilon_t \tag{C.1}$$

Estimating this Taylor approximation using the Ordinary Least Squares (OLS) method enables the testing of linearity under the null hypothesis H_0: $\varphi_2 = \varphi_3 = \varphi_4 = 0$. Rejecting these null hypotheses (indicating a rejection of linearity) suggests that x_t qualifies as a valid transition variable (see Table C.1).

Table C.1
Linearity tests.

	FORSE	AXA
Model 3		
Transition variable	F-stat *P*-value	F-stat *P*-value
R_{t-1}	2.9987e−06	NAN[a]
V_{t-1}	8.1422e−02	8.9482e−91
D_{t-1}	1.1252e−06	1.0243e−116
Model 4		
Transition variable	F-stat *P*-value	F-stat *P*-value
R_{t-1}	4.2616e−07	NAN
V_{t-1}	DEL	6.4202e−96
D_{t-1}	1.5233e−06	NAN
Br_{t-1}	DEL	1.3975e−11
Bv_{t-1}	6.5709e−01	0.0084884
Bd_{t-1}	2.1484e−01	0.15101

[a] If the elements of $x_{(t-d)}$ are close to zero or one, they can lead to invertibility problems. This is because, in the test regression, the powers of the transition variable $x_{(t-d)}$ are included in the regressor. In this case, the output is NAN.

C.2. Specifications of nonlinear models

The STAR model's specification can be determined through a sequence of three F-tests based on the following hypotheses: $H_{0,1}\ :\ \varphi_3 = 0$; $H_{0,2}\ :\ \varphi_2 = 0/\varphi_3 = 0$; $H_{0,3}\ :\ \varphi_1 = 0/\varphi_2 = \varphi_3 = 0$. If the null hypothesis $H_{0,1}$ is rejected, it suggests that the LSTAR model is the preferable choice. However, if $H_{0,2}$ results in a stronger rejection compared to $H_{0,1}$ and $H_{0,3}$, then the ESTAR model is selected. When $H_{0,1}$ and $H_{0,2}$ are not rejected, but $H_{0,3}$ is rejected, it implies that the LSTAR model is the most suitable option. The results of these linearity tests and transition function specifications (LSTAR vs. ESTAR) are presented in Table C.2 below.

Table C.2
Specifications of nonlinear models.

	FORSE	AXA
Model 3		
Transition variable	D_{t-1}	D_{t-1}
$H_{(0,1)} : \phi_3 = 0$	6.8009e−02	0.000953
$H_{(0,2)} : \phi_2 = 0/\phi_3 = 0$	1.8429e−04	6.79e−73
$H_{(0,3)} : \phi_1 = 0/\phi_2 = \phi_3 = 0$	5.3752e−04	2.85e−46
Suggested Model	LSTAR	LSTAR
Model 4		
Transition variable	R_{t-1}	Br_{t-1}
$H_{(0,1)} : \phi_3 = 0$	3.2145e−02	0.000166
$H_{(0,2)} : \phi_2 = 0/\phi_3 = 0$	2.5623e−06	0.013427
$H_{(0,3)} : \phi_1 = 0/\phi_2 = \phi_3 = 0$	2.6672e−02	1.61E−08
Suggested Model	LSTAR	LSTAR

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